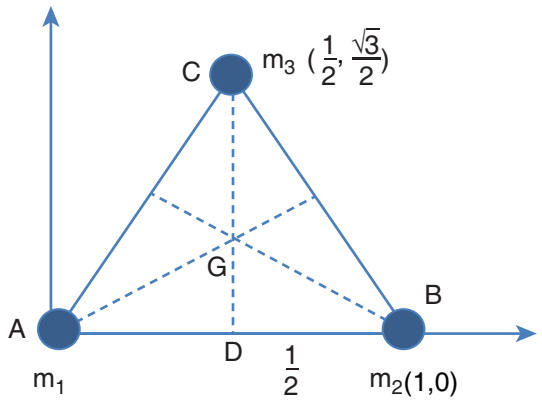
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1m as shown in Figure. Find the position of center of mass.

Solution:



The center of mass of an equilateral triangle lies at its geometrical center G.

The positions of the mass m_1, m_2 and m_3 are at positions A, B and C as shown in the Figure.

From the given position of the masses, the coordinates of the masses m_1 and m_2 are easily marked as (0,0) and (1,0) respectively.

To find the position of m_3 the Pythagoras theorem is applied. As the ΔDBC is a right angle triangle,

$$BC^2 = CD^2 + DB^2$$

$$CD^2 = BC^2 - DB^2$$

$$CD^2 = 1^2 - \left(\frac{1}{2}\right)^2 = 1 - \left(\frac{1}{4}\right) = \frac{3}{4}$$

$$CD = \frac{\sqrt{3}}{2}$$

The position of mass m_3 is

$$\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right) \text{ or } \left(0.5, 0.5\sqrt{3}\right)$$

X Coordinate of center of mass,

$$x_{CM} = \frac{m_1x_1 + m_2x_2 + m_3x_3}{m_1 + m_2 + m_3}$$

$$x_{CM} = \frac{(1\times 0) + (2\times 1) + (3\times 0.5)}{1+2+3} = \frac{3.5}{6}$$

$$x_{CM} = \frac{7}{12}\text{ m}$$

Y Coordinate of center of mass,

$$y_{CM} = \frac{m_1y_1 + m_2y_2 + m_3y_3}{m_1 + m_2 + m_3}$$

$$y_{CM} = \frac{(1\times 0) + (2\times 0) + (3\times 0.5\times \sqrt{3})}{1+2+3} = \frac{1.5\sqrt{3}}{6}$$

$$y_{CM} = \frac{\sqrt{3}}{4}\text{ m.}$$

∴ The coordinates of center of mass G (x_{CM}, y_{CM}) is $\left(\frac{7}{12}, \frac{\sqrt{3}}{4}\right)$

2. An electron of mass $9\times10^{-31} kg$ revolves around a nucleus in a circular orbit of radius 0.53 Å. What is the angular momentum of the electron? (Velocity of electron is, $v = 2.2\times10^{6} m\,s^{-1}$)

Solution:

Mass of the electron, $m = 9\times10^{-31} kg$

Radius of the electron, r = 0.53 Å = $0.53\times10^{-10} m$

Velocity of the electron, $v = 2.2\times10^{6} m\,s^{-1}$

Angular momentum of electron is, $L = I\,\omega$

Electron is considered as a point mass. Hence, its moment of inertia is, $I = m\,r^2$